

Modelling Restoration Policy Outcomes: A Case Study of the Mississippi River Delta

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1. Introduction

The concept of natural capital has gained traction as a viable approach to sustainable resource management. With this shift, a new challenge has emerged in the valuation of natural resources so that they may be accounted for accurately. Much of this difficulty lies in the complexity of the role nature plays in the human economy, and also the various opportunities for subjective interpretation of the contribution natural resources make in daily life. Still, the seeming intractability of this problem should not prevent the attempt of scientific approaches to objectively studying and finding practical methods for natural capital pricing. For several years, researchers and policymakers have been addressing just that through concepts such as payment for environmental/ecosystem services (PES), biodiversity offsets, and carbon markets, to name a few [1-4]. These and similar approaches have, after decades-long persistence, become an integral part of modern policy.

Current computational capacity enables deeper exploration of the complexity introduced by human limits to rationality with the goal of creating more effective metrics, valuations and, eventually, policies for sustainable natural resource consumption. Agent-based models (ABMs) are a widely used tool for studying macro outcomes resulting from micro interactions and heterogeneity [5,6]. Using an ABM, feedbacks between human decisions and ecological-economic processes can be studied in more detail and would garner insight on emergent outcomes at the population level. Such a model would also be useful in strategically developing policy interventions to meet desired outcomes.

2. Suggested scope and research questions

The Mississippi River Delta presents a prime, data-rich case study of competing interests for natural resources in a geographic region prone to natural disasters [7,8]. With respect to the context of disparate multi-industry (oil and gas, fisheries, etc.), landholder, and institutional agendas, complexity-aware tools would provide a distinct advantage in resource valuation and policy development. I propose the construction of an ABM to quantify prices for wetland restoration services, such as blue carbon and coastal protection, and to evaluate market and policy designs (e.g., PES, sediment-diversion offsets). This would start with a single parish as a pilot study to establish core model components, data sources and processing techniques, and calibration methodologies, with plans for eventual extension to the entire delta region.

The aim of the study would be to address spatially targeted sediment feedbacks, impacts on peatland carbon dynamics, heterogeneity in opportunity costs, threshold-avoidance policies, and

price sensitivity to delivery uncertainty by answering the following questions, some related to the particular case and others with potential implications for theory at large:

- Does spatially targeted wetland restoration change local sediment deposition and future suitability for further restoration?
- Do incentives for rewetting alter groundwater/peat oxidation and long-term carbon flux trajectories?
- What is the marginal supply curve for blue-carbon and flood-attenuation services under heterogeneous opportunity costs?
- Are there policy designs that avoid conversion of marshlands to open water while minimizing cost?
- How sensitive are buyer willingness to pay thresholds (and resulting prices) to uncertainty in service delivery attributed to monitoring error and extreme events?

Agent actions would be linked to an empirical proxy for sediment deposition to capture the influence of restoration activities on future restoration suitability through feedback effects, with the option to expand to a process-based model of deposition if time permits. Peatland carbon dynamics would be represented by a peat decomposition module simulating CO₂ and CH₄ fluxes and subsidence; simulations would be sensitive to groundwater levels and driven by agent rewetting decisions. To represent supply curves and market behavior, agent bids will be derived from calculated payoffs. State variables such as elevation relative to mean high water will be tracked and compared against critical value thresholds to identify regime shifts. Uncertainty in monitoring error and storm variability will vary stochastically and be incorporated into buyer beliefs and decisions.

3. Data sources

While data needs remain to be finalized, there are open sources for many of the datasets that would be required for development of the proposed ABM. Relevant remote sensing, hydrology, elevation, and climate data is available from the United States Geological Survey and National Weather Service (9-12). The Louisiana Coastal Protection and Restoration Authority (CPRA) keeps records and reports on restoration projects as well as annual accounts of land use, flood risk, and vegetation change [13]. Field data on CO₂ and CH₄ fluxes is available from the AmeriFlux Network [14]. Plat maps with ownership data could be obtained directly from parish assessors. Most Louisiana-specific blue carbon and PES market transaction data is from smaller studies and trials [13]. If these prove insufficient, general extrapolations could be made from larger market registries. Literature on ecosystem service valuation, PES, market design, risk and uncertainty, and behavioral economics would form the primary basis of agent behaviors and market mechanisms [15-17]. ABMs dealing with variations of the core theoretical concepts of the proposed study have been developed in the past which could also serve as a starting point or inspiration for model mechanics [6].

The proposed work produces a data-driven ABM and supporting analyses that are capable of advancing the practical science of natural capital valuation. The findings would have potential to increase the efficiency and efficacy of sustainable policy and natural resource management in the Mississippi River Delta, and the data integration framework established in the process of model development would define a path for translation to other regions. Another key to maximizing the impact of the project output would be consulting stakeholders in the formation of the model, as it has potential to ultimately serve as a tool for identifying trade-offs among competing interests to increase awareness and support negotiations.

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